



Data trials and triumphs

Introduction

A data analytics professional's job is to provide the data necessary to inform key decisions. They also need to frame their analysis in a way that helps business leaders make the best possible decisions.

In this reading, you're going to explore the role of data in decision-making and the reasons why data analytics professionals are so important to this process. You'll compare data-driven and data-inspired decisions to understand the difference between them. You'll also check out some examples where projects failed or succeeded based on how the data was applied.

Both data-driven and data-inspired approaches are rooted in the idea that data is inherently valuable for making a decision. Well-curated data can provide information to decision-makers that improves the quality of their decisions. Remember: Data does not make decisions, but it does improve them.



Data-driven decisions

As you've been learning, data-driven decision-making means using facts to guide business strategy. The phrase "data-driven decisions" means exactly that: Data is used to arrive at a decision. This approach is limited by the quantity and quality of readily-available data. If the quality and quantity of the data is sufficient, this approach can far improve decision-making. But if the data is insufficient or biased, this can create problems for decision-makers. Potential dangers of relying entirely on data-driven decision-making can include overreliance on historical data, a tendency to ignore qualitative insights, and potential biases in data collection and analysis

Example of a data-driven decision

A/B testing is a simple example of collecting data for data-driven decision-making. For example, a website that sells widgets has an idea for a new website layout they think will result in more people buying widgets. For two weeks, half of their website visitors are directed to the old site; the other half are directed to the new site. After those two weeks, the analyst gathers the data about their website visitors and the number of widgets sold for analysis. This helps the analyst understand which website layout resulted in more widget sales. If the new website performed better in producing widget sales, then the company can confidently make the decision to use the new layout!

Data-inspired decisions

Data-inspired decisions include the same considerations as data-driven decisions while adding another layer of complexity. They create space for people using data to consider a broader range of ideas: drawing on comparisons to related concepts, giving weight to feelings and experiences, and considering other qualities that may be more difficult to measure. Data-inspired decision-making can avoid some of the pitfalls that data-driven decisions might be prone to.

Example of a data-inspired decision

A customer support center gathers customer satisfaction data (often known as a "CSAT" score). They use a simple 1–10 score along with a qualitative description in which the customer describes their experience. The customer support center manager wants to improve customer experience, so they set a goal to improve the CSAT score. They start by analyzing the CSAT scores and reading each of the descriptions from the customers. Additionally, they interview the people working in the customer support center. From there, the manager formulates a strategy and decides what needs to improve the most in order to raise customer satisfaction scores. While the manager certainly relies on the CSAT data in the decision-making process, input of support center representatives and other qualitative information informs the approach as well.



A data analysis triumph

When data is used strategically, businesses can transform and grow their revenue. Consider the example below.

PepsiCo

Since the days of the New Coke launch, things have changed dramatically for **beverage** and other consumer packaged goods (CPG) companies.

According to a *Think with Google* article by Shyam Venugopal, PepsiCo “hired analytical talent and established cross-functional workflows around an infrastructure designed to put consumers’ needs first. Then [the company] set up the right processes to make critical decisions based on data and technology use cases. Finally, [it] invested in the right technology stack and platforms so that data could flow into a central cloud-based hub. This is critical. When data comes together, we develop a holistic understanding of the consumer and their journeys.”

In this data-inspired decision, PepsiCo is not just using its own set of data, but also employing external sources to supplement its datasets and expand its market reach. Learn about how PepsiCo is delivering a more personal and valuable experience to customers using data in [How one of the world’s biggest marketers ripped up its playbook and learned to anticipate intent](#).

Data analysis failures

You’ve been learning why data is such a powerful business tool and how data analysts help their companies make data-driven decisions for great results. Using data to draw accurate conclusions and make good recommendations starts with having complete, correct, and relevant data.

Note: It’s important to remember that it’s possible to have solid data and still make the wrong choices. It’s up to data analysts to interpret the data accurately. When data is interpreted incorrectly, that incorrect interpretation can lead to huge losses. Consider the following.

Coke launch failure

In 1985, New Coke was launched, replacing the classic Coke formula. The company had done taste tests with 200,000 people and found that test subjects preferred the taste of New Coke over Pepsi, which had become a tough competitor. Based on this data alone, classic Coke was taken off the market and replaced with New Coke. The company thought this was the solution to take back the market share that had been lost to Pepsi.

But as it turns out, New Coke was very unpopular—and the company ended up losing tens of millions of dollars. The data seemed correct, but it was incomplete: The data didn't consider how customers would feel about New Coke replacing classic Coke. The company's decision to retire classic Coke was a data-driven decision based on incomplete data.

Mars Orbiter loss

In 1999, NASA lost the \$125 million Mars Climate Orbiter even though the teams had good data. The spacecraft burned to pieces because of poor collaboration and communication. The Orbiter's navigation team was using the International System of Units (newtons) for their force calculations, but the engineers who built the spacecraft used the English Engineering Units system (pounds) for force calculations.

No one realized there was a problem until the Orbiter burst into flames in the Martian atmosphere. Later, a NASA review board investigating the cause of the problem discovered the issue was in the software that controlled the thrusters. One program calculated the thrusters' force in pounds; another program working with the data assumed it was in newtons. The software controllers were making data-driven decisions to adjust the thrust based on 100% accurate data, but these decisions were wrong because of inaccurate assumptions when interpreting it. The two teams might have communicated so they picked a single unit of measure, or so the analysts would have known that conversion was a necessary step in the process to prepare the data. A conversion of the data from one system of measurement to the other could have prevented the loss.

There's a difference between making a decision with incomplete data and making a decision with a small amount of data. You learned that making a decision with incomplete data is dangerous. But sometimes accurate data from a small test can help you make a good decision. Stay tuned: You'll learn about how much data to collect later in the program.

Key takeaways

As a data analyst, you'll rarely need to consider, "Am I being data-driven or data-inspired?" It's helpful to have some context for these two approaches, though your own skills and knowledge will be the most important parts of any analysis project. So, keep a data-driven mindset and ask lots of questions. Experiment with many different possibilities. And use both logic and creativity along the way. Using this approach, you'll be prepared to interpret your data with the highest levels of care and accuracy.